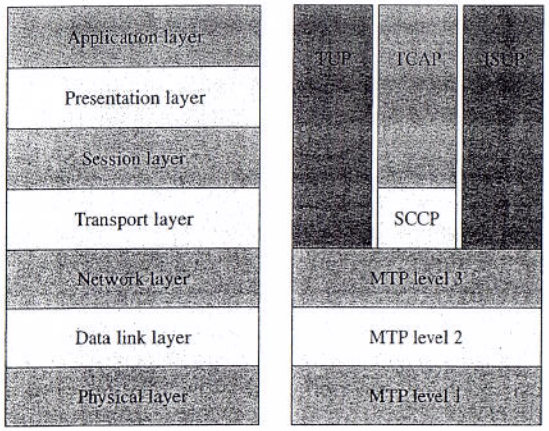

FIGURE 4.54 OSI reference model and SS7 network architecture.

TUP = Telephone user part
TCAP = Transaction capabilities part
ISUP = ISDN user part
SCCP = Signaling connection control part
MTP = Message transfer part

is associated with the setting up, monitoring, and tearing down of connections and belongs to the *control plane*. In the telephone network the user plane is very simple; it consists of physical layer connections for the transfer of voice signals. The control plane of the telephone network is quite sophisticated and involves an entire packet-switching network, the Signaling System #7. The Signaling System #7 (SS7) network is a packet network that controls the setting up, managing, and releasing of telephone calls. The network also provides support for intelligent networks, mobile cellular networks, and ISDN. The SS7 network architecture is shown in Figure 4.54.

This architecture uses "parts" instead of "layers." The message transfer part (MTP) corresponds to the lower three layers of the OSI reference model. Level 1 of the MTP corresponds to the physical layer of the signaling links in the SS7 networks. Physical links have been defined for the following transmission speeds: E-1 (2.048 Mbps = 32 channels at 64 kbps each), T-1 (1.544 Mbps = 24 channels at 64 kbps each), V-35 (64 kbps), DS-0 (64 kbps), and DS-0A (56 kbps). These telephone transmission standards were discussed earlier in this chapter.

MTP level 2 ensures that messages are delivered reliably across a signaling link. This level corresponds to the data link layer in the OSI reference model. The MTP level 3 ensures that messages are delivered between signaling points across the SS7 network. Level 3 provides routing and congestion control that reroutes traffic away from failed links and signaling points.

The ISDN user part (ISUP) protocols perform the basic setup, management, and release of telephone calls. The telephone user part (TUP) is used instead in some countries.

The MTP addresses the signaling points, but is not capable of addressing the various applications that may reside within a signaling point. These applications include

800-call processing, calling-card processing, call management servi
telligent network services. The signaling connection control part (SC
applications to be addressed by building on the MTP to provide cc
connection-oriented service. The SCCP can also translate "global titl
800 number or a mobile subscriber number) into an application iden
tion signaling point. This feature of SCCP is similar to the Internet D
the routing of messages to the appropriate processing point.

The transaction capabilities part (TCAP) defines the messages a
are used to communicate between applications that use the SS7 net
uses the connectionless service provided by the SCCP to support dat
are used in intelligent networks.

TRENDS IN SIGNALING

The notions of signaling and control plane were developed aroun
network, but are currently being adapted and evolved in the contex
Recent work on Internet protocols has developed *generic* signalin
can be used to allocate transmission and switching resources to giv
connections, along a path across a network. The RSVP protocol wa
the establishment of packet connections with service guarantees ac
but the protocol has been extended so that it can be used to set u
across a SONET network as well as lightpaths across an optical netw
this type of signaling in Chapter 10.

A second type of signaling involves the exchange of messag
systems prior to the establishment of a connection. The Session Init
also discussed in Chapter 10, is an application protocol that can be u
modify, and terminate multimedia sessions involving two or more p
can provide the rich set of features of the telephone system, but s
richer and more versatile set of connection types.

◆ 4.7 TRAFFIC AND OVERLOAD CONTROL IN TELEPHONE NETWORKS

In this section we consider the *dynamic* aspects of multiplexing the i
from various users into shared high-speed digital transmission lines. V
ing at the problem of concentration, which involves the sharing of a
by a set of users. Here we examine the problem of ensuring that th
resources, namely, trunks, to provide high availability, that is, low pro
ing. We find that concentration can lead to very efficient usage of n
if the volume of traffic is sufficiently large. We next discuss how this
routing methods in circuit-switching networks. Finally we conside
dealing with overload conditions.

4.7.1 Concentration

Telephone networks are engineered so that, under normal circumstances, there is very little chance that a request for a connection cannot be met. In fact telephone networks make extensive use of concentration to make efficient use of their transmission facilities and to turn a healthy profit from offering telephone service. In this section we explain the statistical basis for the "magic" behind concentration.

Concentration addresses the situation where a large number of users alternate between periods when they need connections and periods when they are idle. In this situation it does not make sense to assign channels to all users all the time. Concentration involves the dynamic sharing of a number of communication channels among a *larger* community of users.

In Figure 4.55 numerous users at a given site, each connected to a local concentrator, share expensive trunks provided by a high-speed digital transmission line to connect to another location, for example, a telephone central office or another user site. When a user on one end wishes to communicate with a user at the other end, the concentrator assigns a communication line or *trunk* for the duration of the call. When the call is completed, the transmission line is returned to the pool that is available to meet new connection requests. Note that signaling between concentrators is required to set up and terminate each call. The number of trunks in use varies randomly over time but is typically much smaller than the total number of lines. We say that a connection request is *blocked* when no trunks are available. The objective here is to minimize the number of trunks, while keeping the probability of blocking to some specified level.

Figure 4.56 shows the occupancy of a set of seven trunks over time. The shaded rectangles indicate periods when a given trunk is in use. The upper part of the figure shows the corresponding $N(t)$, the number of trunks in use at time t. In this example the system is in a blocking state when $N(t) = 7$.

The users require trunk connections in a sporadic and unscheduled manner. Nevertheless, the statistical behavior of the users can be characterized. In particular it has been found that the users *as a group* make requests for connections according to a Poisson process with connection request or **arrival rate** λ calls/second. A Poisson process is characterized by the following two properties:

1. In a very small time interval of duration Δ seconds, only two things can happen: There is a request for one call, with probability $\lambda\Delta$, or there are no requests for calls, with probability $1 - \lambda\Delta$.
2. The arrivals of connection requests in different intervals are statistically independent.

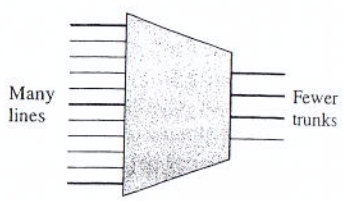

FIGURE 4.55 Concentration (bold lines indicate lines/trunks that are in use).

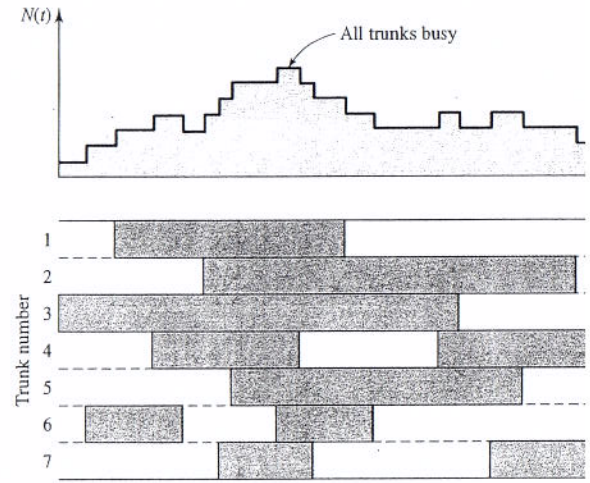

FIGURE 4.56 Number of trunks in use as a function of time.

The first property states that the rate at which call requests are mad time. For example, it does not depend on the number of users already second property states that the pattern of prior requests does not influen of a request occurring in a given time interval. Both of these propertie when dealing with the aggregate behavior of a large number of peo of the trunk concentration problem is carried out in Appendix A. We results of the analysis here.

The time that a user maintains a connection is called the **holding** the holding time X is a random variable. The average holding time $E[$ as the amount of "work" that the transmission system has to do for telephone systems typical conversations have a mean holding time of The **offered load** a is defined as the total rate at which work is offered b of users to the multiplexing system, as measured in Erlangs:

$$a = \lambda \times E[X] \qquad \text{(Erlangs)}$$

One Erlang corresponds to an offered load that would occupy a sing cent of the time, for example, an arrival rate of $\lambda = 1$ call/second a time of $E[X] = 1$ second/call would occupy a single trunk all of th telephone systems are designed to provide a certain level or "grade" the *busy hour* of the day. Measurements of call attempts reveal clear p and relatively stable patterns of call attempts. For example, in office rate peaks in mid-morning, drops during lunch hour, and peaks again i In the subsequent discussion the offered load should be interpreted a the busy hour.

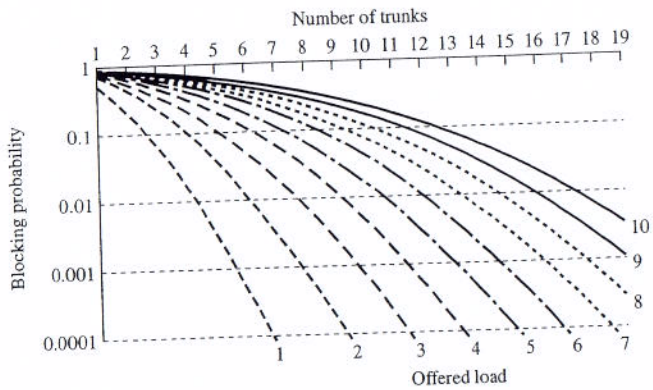

FIGURE 4.57 Blocking probability for offered load versus number of trunks: Blocking probability decreases with the number of trunks.

The blocking probability P_b for a system with c trunks and offered load a is given by the **Erlang B formula:**

$$P_b = B(c, a) = \frac{a^c/c!}{\sum_{k=0}^{c} a^k/k!} \tag{4.6}$$

where $k! = 1 \times 2 \times 3 \ldots \times (k-1) \times k$.

Figure 4.57 shows the blocking probability for various offered loads as the number of trunks c is increased. As expected, the blocking probability decreases with the number of trunks. A 1% blocking probability is typical in the design of trunk systems. Thus from the figure we can see that four trunks are required to achieve this P_b requirement when the offered load is one Erlang. On the other hand, only 16 trunks are required for an offered load of nine Erlangs. This result shows that the system becomes more efficient as the size of the system increases, in terms of offered load. The efficiency can be measured by trunk **utilization** that is defined as the average number of trunks in use divided by the total number of trunks. The utilization is given by

$$\text{Utilization} = \lambda(1 - P_b)E[X]/c = (1 - P_b)a/c \tag{4.7}$$

Table 4.2 shows the trunk utilization for the various offered loads and $P_b = 0.01$. Note that for small loads the utilization is relatively low. In this case extra trunks are required to deal with surges in connection requests. However, the utilization increases as the size of the systems increases in terms of offered load. For a load of two Erlangs, a total of 7 trunks is required; however, if the load is tripled to six Erlangs, the number of trunks required, 13, is less than double. The entry in Table 4.2 for offered loads of 50 and 100 Erlangs shows that high utilization is possible when the offered loads are large. These examples demonstrate how the sharing of network resources becomes more efficient as the scale or size of the system increases. The improvement in system performance that results from aggregating traffic flow is called *multiplexing gain*.

Telephone networks have traditionally been engineered for probability of blocking of 1% during the busy hour of the day. The actual call request rates at other times of

TABLE 4.2 Trunk utilization.

Load	Trunks @ 1%	Utilization
1	5	0.20
2	7	0.29
3	8	0.38
4	10	0.40
5	11	0.45
6	13	0.46
7	14	0.50
8	15	0.53
9	17	0.53
10	18	0.56
30	42	0.71
50	64	0.78
60	75	0.80
90	106	0.85
100	117	0.85

the day can be much lower, so the probability of blocking is even lc have come to expect that the network is always available to complet

4.7.2 Routing Control

Routing control refers to the procedures for assigning paths in a n tions. Clearly, connections should follow the most direct route, si uses the fewest network resources. However, we saw in Section 4.7. flows are not large the required set of resources to provide high ava blocking probability of 1%, will be used inefficiently. Economic con an approach that provides direct trunks between switches that have between them and that provide indirect paths through tandem sw flows.

A hierarchical approach to routing is desirable when the volume switches is small. This approach entails aggregating traffic flows c shared by multiple switches. Consider the situation in Figure 4.58 B, and C have 10 Erlangs of traffic to D, E, and F. Suppose that swi are close to each other and that they have access to tandem switch 1.

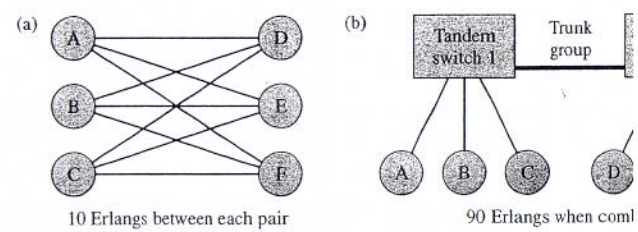

FIGURE 4.58 Hierarchical routing control.